

# Graph Neural Networks for Defect Localization in Two-Layer CFRP from Ultrasonic NDT

Architecture study under a balanced noise-augmented recipe

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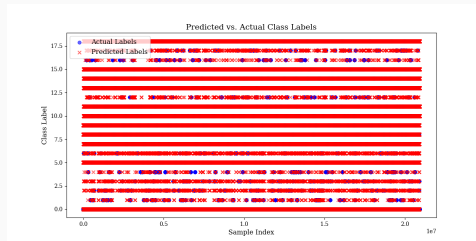
# Motivation: NDT defect localization in layered CFRP

Carbon-fibre laminates around bolt holes accumulate **sub-surface delaminations** invisible to the eye.

Ultrasonic NDT yields a per-node response field on a **two-layer mesh** — but reading it by hand is slow and operator-dependent.

## Goal

Predict, for every mesh node, *which of 19 defect classes* (incl. “defect-free”) it belongs to — directly from the mesh.



Predicted vs. true node labels on the hole mesh.

## One specimen $\rightarrow$ one graph

- 13,942 nodes (two stacked layers around the hole)
- Node features:  $[x, y, z, \text{DSPSS}]$
- Edges: FEM mesh connectivity (+ coordinate-difference edge features  $[dx, dy, dz, \|\cdot\|]$ )
- 19 classes; defect-free is class 0

**Input** = *noise-augmented difference* field (damaged – pristine), which isolates the defect signal.

## The trap: extreme imbalance

Class-0 prior  $\approx 0.998$ . A model that predicts “defect-free” everywhere scores 99.8% accuracy and **macro-F1  $\approx 0$** .

$\Rightarrow$  we report **macro-F1** and engineer the training distribution, not just the loss.

# The recipe: balance the data, not (only) the loss

1. **Noise-augmented difference data** — robustness to acquisition noise.
2. **Defect-free negatives (NDF)** — 5000 genuine class-0 graphs added to *train only*.
3. **Balanced sampling** — `-defect_cap 5000 -ndf_count 5000`  $\rightarrow$  5000:5000.
4. Leakage control: group-purged val/test split by specimen.

## Key finding

With the recipe in place, the *data distribution dominates*: a faithful reproduction reaches **macro-F1**  $\approx 0.70$ , vs.  $\leq 0.08$  without it.

# Architecture sweep under the balanced recipe

Same recipe, 120 epochs, balanced 5000:5000, group-purged eval.

| Architecture             | macro-F1     | note                                  |
|--------------------------|--------------|---------------------------------------|
| <b>MeshGraphNet</b>      | <b>0.764</b> | Encode-Process-Decode, edge update    |
| GATv2                    | 0.694        | dynamic attention                     |
| GAT (Frontiers baseline) | 0.674        | prior published model ( $\sim 0.61$ ) |
| GraphSAGE                | 0.663        | max-aggregation                       |
| PNA                      | 0.613        | multi-aggregator                      |

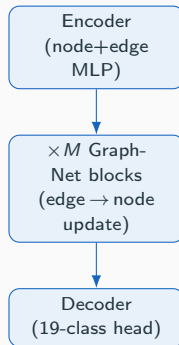
## Takeaway

Under the balanced recipe, the **mesh-physics architecture (MeshGraphNet)** wins — beating the legacy GAT recipe (0.70) and the published baseline (0.61).

# Why MeshGraphNet fits the problem

## Encode–Process–Decode (Pfaff et al., 2021):

- node *and* edge embeddings updated at *every* message-passing step (residual);
- edge feature = coordinate difference  $[dx, dy, dz, \text{dist}] \Rightarrow$  local geometry & anisotropy;
- matches FEM-mesh stress-propagation structure better than attention-only GNNs.



## Novelty: four MeshGraphNet variants for layered NDT

| Variant                | Idea / why it could matter for CFRP                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MeshGraphKAN</b>    | Fourier-KAN node encoder — learnable spectral basis for the <i>high-frequency stress-concentration</i> field (spectral-bias mitigation). |
| <b>Hybrid-MGN</b>      | mesh edges + <i>front/back cross-layer “world” edges</i> — structurally separates the two laminate layers (no prior work).               |
| <b>BiStride-MGN</b>    | per-graph global-context broadcast — multiscale, long-range stress propagation (under-reaching).                                         |
| <b>MGN-Transformer</b> | interleaved per-graph self-attention — global node interaction.                                                                          |

### Hypothesis

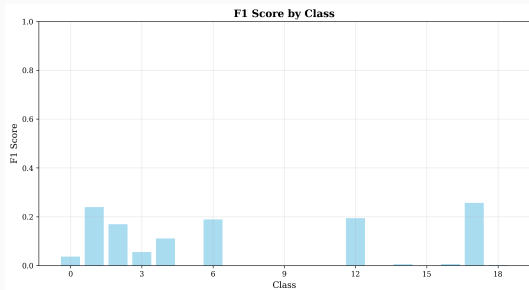
**MeshGraphKAN** × **stress concentration** and **Hybrid-MGN** × **two-layer separation** are the two novel angles most likely to push past 0.80.

| Variant                 | macro-F1 | vs. MeshGraphNet (0.764) |
|-------------------------|----------|--------------------------|
| MeshGraphNet (baseline) | 0.764    | —                        |
| MeshGraphKAN            | TBD      |                          |
| Hybrid-MGN              | TBD      |                          |
| BiStride-MGN            | TBD      |                          |
| MGN-Transformer         | TBD      |                          |

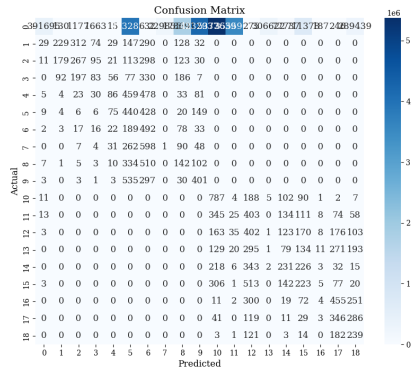
## Target

Cross the **0.80** macro-F1 line with a physically-motivated variant.

# Per-class performance & confusion



Per-class F1.



Confusion matrix.

- The **data recipe** (noise-aug difference + NDF negatives + balancing) is the dominant factor: 0.08  $\rightarrow$  0.70.
- Under the balanced recipe, **MeshGraphNet** is the best architecture (**0.764**), beating the published GAT baseline (0.61).
- Two **novel layered-NDT variants** — Fourier-KAN and front/back Hybrid edges — target  $> 0.80$ .

## Future work

- **Application:** transfer to aerospace payload-fairing SHM (H3 / GNN-SHM).
- **Validation:** incorporate experimental (non-simulated) NDT data.

Thank you.

Questions?